TCS NQT Pattern

Section	Duration (in minutes)
Part A – Foundation Section	**75**
Numerical Ability	25
Verbal Ability	25
Reasoning Ability	25
Part B – Advanced Section	**115**
Advanced Quantitative and Reasoning Ability	25
Advanced Coding(2 - 3 Qs)	90
Total Duration	**190**

26 April – Exam Experience Summary

Point No.	Exam Feature
1	No question-wise timer (only section-wise timer)
2	Sectional and overall cut-off applicable
3	On-screen calculator available
4	Rough paper allowed
5	No negative marking
6	Section switching not allowed
7	Revisit questions – not allowed

Difficulty Level – Foundation Section

Section	Difficulty Level
Numerical Ability	Medium to Difficult
Verbal Ability	Easy to Medium
Reasoning Ability	Easy to Medium

Difficulty Level – Advanced Section

Section	Difficulty Level / Details
Advanced Quantitative & Reasoning	Medium to Difficult
Advanced Coding – Question 1	Easy to Medium (Array-based)
Advanced Coding – Question 2	Medium (Matrix-based)

Detailed Topic Breakdown

1 Numerical Ability

Category	Topics Covered
Number System	Number System, LCM & HCF, Divisibility, Numbers & Decimal Fractions
Mensuration	Geometry, Area, Shapes & Perimeter
Arithmetic Ability	Percentages, Allegations & Mixtures, Ratios & Proportion, Work & Time, Speed–Time–Distance, Profit & Loss, Averages, Equations, Ages, Probability, Clocks & Calendar
Elementary Statistics	Mean, Median, Mode, Variance, Standard Deviation
Data Interpretation	Pie Charts, Tabular DI, Graphical DI
Simplification & Approximation	Simplifications and Calculations (Rational & Irrational Numbers)

TCS NQT – Important Quant Topics (Weightage Oriented)

Topic	Expected Questions
Simplification	2–3
Percentage	Moderate
Data Interpretation	1 diagram (2–3 questions)
Ratio & Proportion	1–2
Profit & Loss, Time & Work	Moderate
Simple & Compound Interest	Moderate
Time, Speed & Distance	Trains, Boats (2–3 questions)

2 Verbal Ability – Topic Breakdown

Area	Topics
Vocabulary	Antonym, Synonym, Meanings
Grammar	Error Correction, Tenses, Conjunctions, Prepositions
Sentence Skills	Word Completion, Sentence Completion, Sentence Rearrangement
Structure & Flow	Formal / Informal Sentences, Sentence Joining
Reading Skills	Reading Comprehension
Error Handling	Error Identification

3 Reasoning Ability – Detailed Topic Breakdown

Category	Topics Covered
Pattern Identification	Word Pattern, Letter Series
Numeric Patterns	Number Series
Problem Solving	Seating Arrangements, Blood Relations, Data Arrangements
Decision Making	Decision Making, Syllogism, Data Sufficiency
Propositional Reasoning	Statement-based Questions, Logical Venn Diagrams
Visual / Spatial Reasoning	Cube Folding, Paper Cuts, Folds, Visual & Spatial Reasoning

Reasoning – High-Frequency Topics (Exam Focus)

Topic
Number / Alphanumeric Series
Order & Ranking
Figure-Based Problems
Seating Arrangement
Syllogism
Direction Sense
Blood Relation
Coding & Decoding

4 Advanced Quantitative Aptitude (Advanced Section)

Topics
Puzzles
Averages
Profit & Loss
HCF & LCM
Permutation & Combination
Probability
Geometry
Mensuration
Partnership
Number System

Time & Work
Simple & Compound Interest
Progressions
Logarithms

5 Advanced Reasoning Ability (Advanced Section)

Topics
Data Arrangements
Visual Reasoning
Venn Diagrams
Syllogisms
Statement & Assumption
Statement & Conclusion
Seating Arrangements
Spatial Reasoning
Puzzles
Cuts, Folds & Cubes-based Questions

6 Advanced Coding – Exam Pattern

Component	Details
Number of Questions	2
Duration	90 minutes
Difficulty	Easy–Medium (Q1), Medium (Q2)
Common Focus	Arrays (Q1), Matrix (Q2)

7 Top Coding Topics for TCS NQT

Priority	Topic
1	Basic Maths
2	Arrays
3	Strings

4	Sorting
5	Searching
6	Matrix
7	Collections / STL
8	Recursion
9	Dynamic Programming
10	Number System

ADVANCE CODING PROBLEM SET - 1

Arrays & Strings (Logic + Constraints)

Problem	LC Difficulty	PRIME Tag
Two Sum	Easy	Foundation (logic clarity)
Best Time to Buy and Sell Stock	Easy	Optimization thinking
Product of Array Except Self	Medium	Prefix/Suffix
Longest Substring Without Repeating Characters	Medium	Sliding Window
Container With Most Water	Medium	Two Pointers
Minimum Window Substring	Hard	PRIME Core

Hashing & Frequency Logic

Problem	LC Difficulty	PRIME Tag
Valid Anagram	Easy	Frequency basics
Subarray Sum Equals K	Medium	Prefix Sum + Hash
Longest Consecutive Sequence	Medium	Optimization
Group Anagrams	Medium	Mapping logic
First Missing Positive	Hard	Index Mapping (PRIME fav)

Recursion & Backtracking (Thinking Test)

Problem	LC Difficulty	PRIME Tag
Fibonacci Number	Easy	Recursion base
Subsets	Medium	Decision Tree
Permutations	Medium	State Tracking
Combination Sum	Medium	Constraint pruning
N-Queens	Hard	PRIME Benchmark

Linked List (Pointer Reasoning)

Problem	LC Difficulty	PRIME Tag
Reverse Linked List	Easy	Pointer basics
Linked List Cycle	Easy	Floyd's logic
Remove Nth Node From End	Medium	Two Pointers
Add Two Numbers	Medium	Simulation
Reverse Nodes in k-Group	Hard	Pointer Mastery

Stack & Queue (Monotonic Thinking)

Problem	LC Difficulty	PRIME Tag
Valid Parentheses	Easy	Stack intro
Min Stack	Medium	Auxiliary structure
Daily Temperatures	Medium	Monotonic Stack
Next Greater Element II	Medium	Circular logic
Largest Rectangle in Histogram	Hard	PRIME Stack Core

Trees & BST (Recursive Depth)

Problem	LC Difficulty	PRIME Tag
Maximum Depth of Binary Tree	Easy	Traversal
Diameter of Binary Tree	Medium	Post-order logic
Validate Binary Search Tree	Medium	Range reasoning
Lowest Common Ancestor	Medium	Recursive thinking
Binary Tree Maximum Path Sum	Hard	PRIME Tree Core

Graphs (PRIME Intro Level)

Problem	LC Difficulty	PRIME Tag
Number of Islands	Medium	DFS/BFS
Flood Fill	Easy	Grid traversal

Clone Graph	Medium	Graph mapping
Course Schedule	Medium	Cycle detection
Word Ladder	Hard	PRIME BFS Benchmark

Dynamic Programming (Controlled PRIME)

Problem	LC Difficulty	PRIME Tag
Climbing Stairs	Easy	DP intro
House Robber	Medium	1D DP
Coin Change	Medium	Optimization DP
Longest Increasing Subsequence	Medium	DP + Binary Search
Edit Distance	Hard	PRIME DP Core

Greedy (Proof-Based Thinking)

Problem	LC Difficulty	PRIME Tag
Assign Cookies	Easy	Greedy intro
Jump Game	Medium	Reachability
Gas Station	Medium	Circular logic
Non-overlapping Intervals	Medium	Interval greedy
Candy	Hard	PRIME Greedy Logic

ADVANCE CODING PROBLEM SET - 2

(Less popular, high-thinking, PRIME only)

Arrays & Prefix / Window (PRIME Logic)

Problem	LC Difficulty	Why PRIME
Subarray Sums Divisible by K	Medium	Prefix + modulo reasoning
Continuous Subarray Sum	Medium	Mathematical constraint logic
Find the Duplicate Number	Medium	Cycle detection on array
Shortest Subarray with Sum at Least K	Hard	Deque + prefix optimization
Maximum Sum of Two Non-Overlapping Subarrays	Hard	Multi-window DP thinking

Math + Number Theory (PRIME Favorite)

Problem	LC Difficulty	Why PRIME
Count Primes	Medium	Optimization, not brute
Ugly Number II	Medium	Multiple pointer logic
Reordered Power of 2	Medium	Digit hashing
Minimum Moves to Equal Array Elements II	Medium	Median logic
Poor Pigs	Hard	Logical math modeling

Recursion / Backtracking (Non-obvious)

Problem	LC Difficulty	Why PRIME
Different Ways to Add Parentheses	Medium	Divide & conquer recursion
Beautiful Arrangement	Medium	State pruning
Palindrome Partitioning II	Hard	Backtracking + DP
Expression Add Operators	Hard	State explosion control
Remove Invalid Parentheses	Hard	BFS + pruning

Linked List / Pointer Intelligence

Problem	LC Difficulty	Why PRIME

Reorder List	Medium	Pointer manipulation
Linked List Random Node	Medium	Reservoir sampling
Split Linked List in Parts	Medium	Constraint distribution
Copy List with Random Pointer	Medium	Hash + pointer mapping
Sort List	Hard	Merge sort on LL

Stack / Monotonic / Interval Logic

Problem	LC Difficulty	Why PRIME
Remove K Digits	Medium	Greedy + stack
Asteroid Collision	Medium	Simulation correctness
Maximal Rectangle	Hard	Histogram + DP combo
Sum of Subarray Minimums	Medium	Monotonic stack math
Exclusive Time of Functions	Medium	Stack + timestamps

Trees (Reasoning > Traversal)

Problem	LC Difficulty	Why PRIME
Construct Binary Tree from Preorder and Inorder	Medium	Index mapping
Count Complete Tree Nodes	Medium	Binary search on tree
Serialize and Deserialize Binary Tree	Hard	Design + recursion
All Nodes Distance K in Binary Tree	Medium	Graph thinking
Binary Tree Cameras	Hard	Greedy on trees

Graphs (PRIME Core)

Problem	LC Difficulty	Why PRIME
Is Graph Bipartite?	Medium	Coloring logic
Minimum Height Trees	Medium	Topological pruning

Shortest Path in Binary Matrix	Medium	BFS with constraints
Evaluate Division	Medium	Graph modeling
Critical Connections in a Network	Hard	Tarjan intuition

Dynamic Programming (PRIME Style)

Problem	LC Difficulty	Why PRIME
Partition Equal Subset Sum	Medium	Subset DP
Target Sum	Medium	DP state transformation
Ones and Zeroes	Medium	2D DP
Burst Balloons	Hard	Interval DP
Palindrome Partitioning III	Hard	DP + optimization

Greedy / Design Thinking

Problem	LC Difficulty	Why PRIME
Queue Reconstruction by Height	Medium	Sorting + placement
Task Scheduler	Medium	Frequency logic
Minimum Number of Arrows to Burst Balloons	Medium	Interval greedy
Car Pooling	Medium	Difference array
Meeting Rooms III	Hard	Simulation + heap